

1/6/20

$$\frac{6}{100} = \frac{24}{100} + \frac{1}{100} + \frac{3}{100} + \frac{1}{100}$$

$$AB(2,1) + ABC + ABC + (AB) + A(BA) (=4)$$

$$PDC + AC = \frac{1}{2}PBC + \frac{1}{2}ABC + \frac{1}{2}ABD + \frac{1}{2}ADC + \frac{1}{2}PBC + \frac{1}{2}PDC$$

$$\overline{PQ} \cong \overline{PR} \cong \overline{PS} \cong \overline{PT}$$

$$5.0 \times 10^{-2} + 0.25 + 0.001 + 0.0001$$

A.B.C. +

$$h(\bar{g}) = h(\bar{g} \cap \bar{h}) + h(\bar{g} \cap \bar{h}^c)$$

陳士英 著

$$(B_n(t))_{n \in \mathbb{N}} \subset (T_n + B_n(t))_{n \in \mathbb{N}}$$

0.4 W. 4. 100

$$B_0 = (B_0^1, B_0^2, B_0^3)$$

6. 4. 1968

$$y = (1 + \frac{1}{2}x)^{1/2}$$

$\text{O}_2 + \text{P} \rightarrow \text{PO}_2$
 even binding with strong acid
 oxygen strong acid

$\mathbb{H} \rightarrow G$

6-13

1015

1070

101102

1997-1998

James May, *Life in May*

1. *Journal of the American Medical Association*, 1997; 278: 1039-1044.

1992

44 45

control

0191

1994

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 26

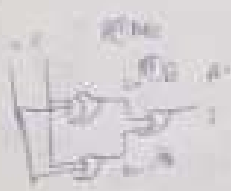


Figure 1

1968

THE UNIVERSITY OF CHICAGO

1990

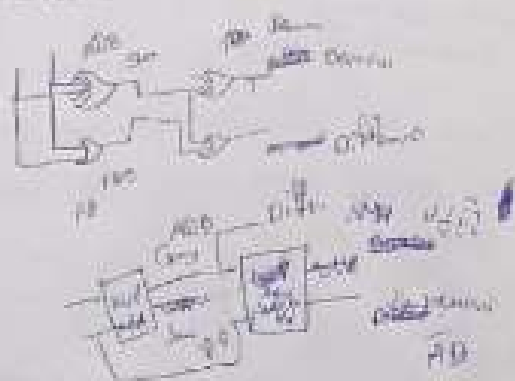
Am. J. Hyg. 57: 333-344, 1948

340 J. Polym. Sci. Part A: Polym. Chem.: Vol. 42, 337-343 (2004)

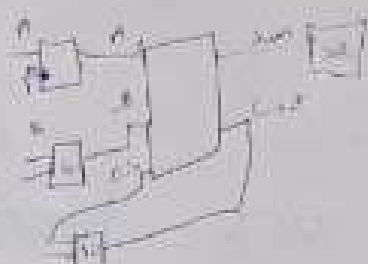
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$$L(\varphi(\bar{x})) = (\bar{a}, \bar{b}) \models \varphi$$

Teil 1: 2 Halbkreis

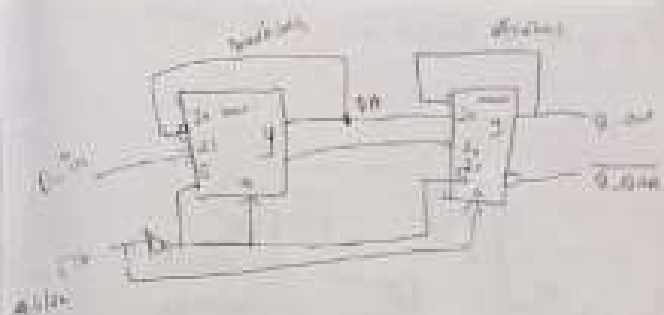


2. Teil: 2 Halbkreis



3. Teil: 2 Halbkreis

gibt es



4. Teil: 2 Halbkreis



5. Teil: 2 Halbkreis

2. Teil: 2 Halbkreis

S	R	Q
0	1	0
1	0	1
0	0	1
1	1	0

→ Wahrheitstabelle

S.R. K.

Summe

→ also 2. Teil

3. Teil: 2 Halbkreis

4. Teil: 2 Halbkreis

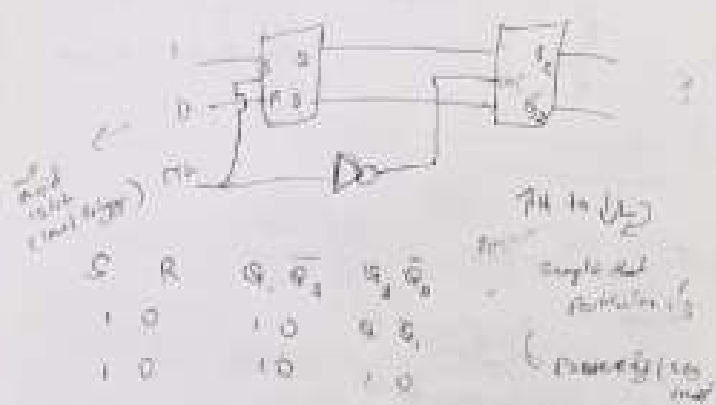
5. Teil: 2 Halbkreis

6. Teil: 2 Halbkreis

7. Teil: 2 Halbkreis

gibt es





S	R	Q ₁	Q ₂	Q ₁ Q ₂
1	0	1	0	10
1	0	1	0	10

Q ₁	Q ₂	Q ₁ Q ₂
0	0	00
0	1	01
1	0	10
1	1	11

Q ₁	Q ₂	Q ₁ Q ₂
0	0	00
0	1	01
1	0	10
1	1	11

Q_{out} = D 1000
 In flip-flop

Q ₁	Q ₂
0	0
0	1
1	0
1	1

Transfer to Q_{out}, 1000



multiple delay
 weight delay
 system delay
 delay two cycle
 100 1/2

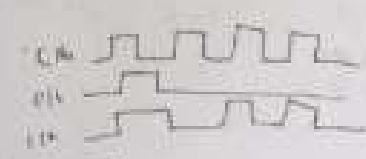


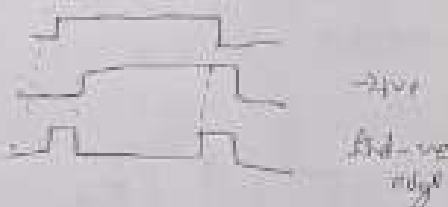
100 delay

multiple delay
 reg 1, 16
 reg 16, 16



Delay
 Reg like
 without feedback, reverse





prof. C. Day - completed system
in last week

Presented by the author, 1978

Page 1 of 1

1947

Pharmaceutical D. H. H.

9	9	2	4
7	0	3	7
0	1	1	7
1	0	7	0
1	1	7	0

$$\mathcal{D} \quad \mathcal{Q}_m \quad \mathcal{Q}_{m+1}$$

Q	D	O
Q	I	0
I	O	*
I	I	I

$$Q_{\text{int}} = J_{\text{G}} + R_{\text{G}}$$
$$= 76.41$$
$$\frac{dy}{dx} = -\frac{y}{x}$$

— 10 —

4. Public Interest - The public interest is the interest of the community as a whole. It is the interest of the community in the well-being of its members and in the maintenance of the public order and the public peace.

W. C. A. R. D.

English → multiple Q

6. $\frac{d}{dx} \ln x = \frac{1}{x}$

Calceolaria *gambelii*

$$\rho_1 \rho_2 \rightarrow \rho_1 \rho_2 \text{ Sch.}$$

1999

المجلس الأعلى للدراسات الإسلامية

Let $\mathcal{A} = \{A_1, \dots, A_n\}$ be a family of n sets.

Syllable - small letter
Cursive

Handwritten: $\{ \text{Handwritten} \rightarrow \text{not applicable} \}$

وَلَمْ يَكُنْ لَهُ كُفُوًا عَلَيْهِ سَكَتٌ

Buss: 2 phas → up/down / 1/2 scale
 ↓
 ↓ downed line

more & more

tasks → synthesis block → not
 → stimulate
 → verify

Tasks → Program → synthesis → H.W.

FSM → HDLC ⇒ 7E
 strong values
 0111110

S/O/O



3k → OFF

out
 1 = T/F
 0 = 1/0 or a/b

many @ (highly in) → still
 change

out

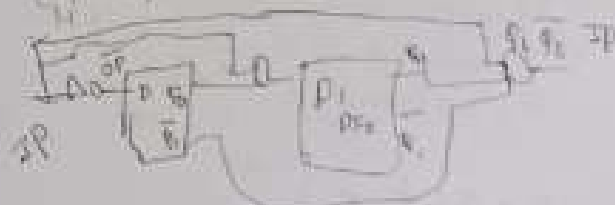
wire → 7-BH impedance (2)
 100 = x

3k → 1000
 out } error



Counting → up-down → H.W.
 30 → 0 → 30

Input	Output	Input	Output	Input	Output
00	0	01	0	10	0
01	1	10	1	11	1
10	0	11	0	00	1
11	1	00	1	01	0

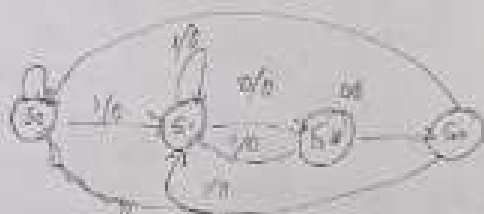


V.H.N. using code

2

2012

7. $\frac{1}{2} \log \frac{1}{2}$ and $\frac{1}{2} \log \frac{1}{2}$



date	place	alt	wind	temp	precip	humidity
1-1	0	0	0	0	0	0
1-2	0	0	0	0	0	0
1-3	0	0	0	0	0	0
1-4	0	0	0	0	0	0
1-5	0	0	0	0	0	0
1-6	0	0	0	0	0	0
1-7	0	0	0	0	0	0
1-8	0	0	0	0	0	0
1-9	0	0	0	0	0	0
1-10	0	0	0	0	0	0

$$y \cdot \partial_x = \partial_x (y)$$

[illegible]

$$C_0 = 9.97 \times 10^{-10}$$

96

507

1144-4259

Profoundly thankful

1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 26

18. $\frac{1}{2} \pi, \frac{3}{2} \pi$ and $\frac{5}{2} \pi$.

substantive value to bond in bond.

$$I = \frac{1}{2} \pi \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} = \frac{1}{4} \pi$$

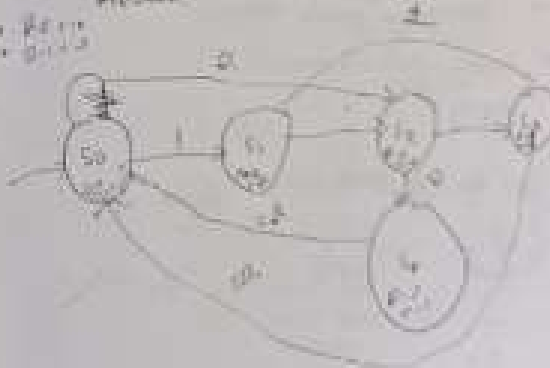
1 x 5000 = 5000

$\frac{50000}{100} = 500$

25th Nov

100

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10/6/50

ASIC flow
 high standards
 RTL code
 functional blocks
 logic gates
 logic gates
 0-1 logic = hardware blocks

locked against other things - hardware (physical)
 physical constraints - logic, layers



FIFO → Data in

how often

Expensive how to give address
 Read expensive write cheap
 empty

write & read



Depth = 16

100, 100, 2

11/6/50

DSP430E

Expensive in FPGAs

Pr. J. H.

50-70

$$\frac{100}{10} = 10$$

area - cost 100

cost 100

100

100

100

100

block memory in FPGAs

1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100.

BRAM → embedded memory
 512 BRAM

1000

1000

1000

1000

1000

1000

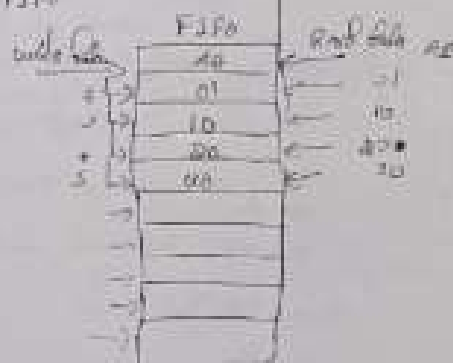


② Dual Port BRAM (100000)

12 = 2 independent ports A & B

• Different addresses and same clk

③ FIFO



1 byte = 8 bits

1 word = 32 bits

1 nibble = 4 bits

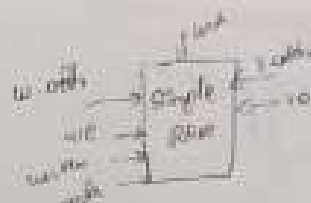
13/6/21 write - synchronous

read - Asynchronous to synchronous

memory → 8000 bits per hour

1 port for each read & write

• Config dual port from



bandwidth

① FA > FB



FIFO depth calculation

write = 1000

read = 5000

$\frac{1}{1000} = 10 \mu s$

$\frac{1}{5000} = 2 \mu s$

$\frac{1}{5000} = 2 \mu s \rightarrow$ read rate

$\frac{15000}{2000} = 7.5 \rightarrow$ 7.5 words

burst length → no. of data within 1 burst

remaining no. of data to be read = 15000 - 12000 = 3000

② FA > FB

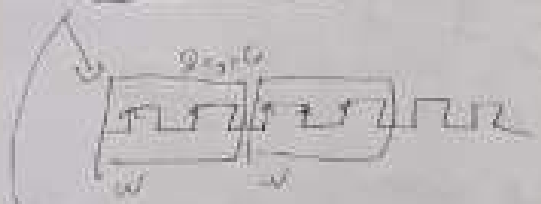
FA = 10000, FB = 20000

burst length = 100

no. of data to be read = 10000

no. of data to be read = 10000

q cycle



$$T = \frac{1}{50000} = 0.5 \mu s \Rightarrow 10$$

$$or \frac{1}{50000} = 0.5 \mu s \Rightarrow 10$$

$$100 \times 0.5 = 50000$$

write the data out

$$30000 \times 1/50 = 600 \rightarrow \text{no. of data words}$$

$$\text{cycle time} = 1/50 = 120 \times 32 = 3840$$

③ $FA < FB$

idle cycle $\rightarrow$ on cycle data on empty cycle

④ $FA < 1A$

$$W = 30000 \rightarrow 1A$$

$$R = 50000 \rightarrow 5B$$

2. $V_{max} = 60000$ idle

$$W = 60000 \rightarrow 30000$$

$$\text{bus length} = 20$$

$$n \text{ cycle} = W = 1$$

$$n \text{ cycle} = R = 3$$

$$120 = 100 = 20$$

$$R = 9 \times 10^4 = 90000$$

$$\frac{90000}{20} = 4500$$

$$T = 10 \times \frac{1}{50} = 90$$

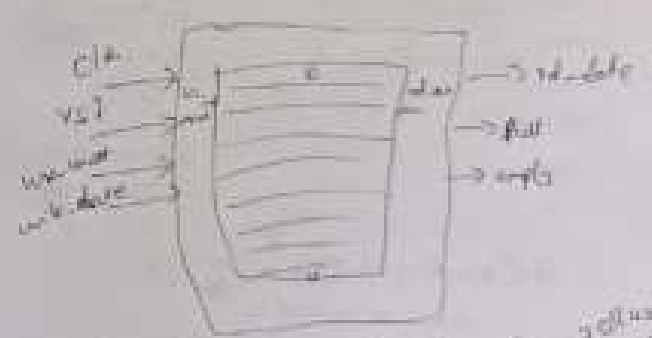
$$120 \times 10 = 1200$$

$$4 \times \frac{1}{50} = 4000 = 2000 \text{ words}$$

$$\frac{40000}{20} = 2000$$

$$120 = 100 = 20$$

write the



write & full out can't read $\rightarrow$ full
both read & write to equal $\rightarrow$ empty

11/1/20

$$\frac{P_1}{P_2} = 1$$

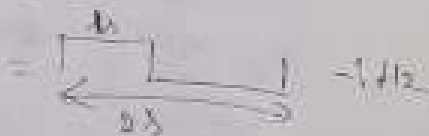
$$\frac{Q_{P1}}{Q_{P2}} = 1$$

12 M

$$= \frac{24M+2}{24M+2}$$

$$SE = 11, 5.99$$

$$= 1$$



$$= \frac{24M+2}{12M+2}$$

$$2M = 2M$$



$$= \frac{24M+2}{2 \times 2M+2}$$

$$= \frac{24M+2}{4M+2}$$

M

$$= 1, 0.00, 0.00$$

$$5, 99.9, 99.9$$

$$if (Q_{P1} = 1, 7.99)$$

$$Q_{P2} = 1, 7.99$$

$$if (Q_{P1} = 5.99, 99.9)$$

$$Q_{P2} = 5.99, 99.9$$



$$\frac{Q_{P1}}{Q_{P2}}$$

Q_{P1}

if 12 M value small

Q_{P1} & Q_{P2} value small

and value of P₁ & P₂

$$\frac{Q_{P1}}{Q_{P2}}$$

Q

-3

$$2, 000, 000$$

$$\frac{Q_{P1}}{Q_{P2}}$$

Q_{P1}

$$\frac{Q_{P1}}{Q_{P2}}$$

Q

$$1, 500, 000$$

Q_{P1}

Q_{P2}

Q_{P1}

Q_{P2}

11/1/20

on the 11/1/20

Q_{P1} -> value of P₁ & P₂

Q_{P1} & Q_{P2}

Q_{P1} & Q_{P2} value small

Q_{P1} & Q_{P2} value small

Communication Protocol

Q_{P1} & Q_{P2}

$$100M \sim 10 \times 10^6 \sim 0.1 \times 10^8$$

$$\frac{100M}{10^8}$$

$$\frac{1000000}{1000000}$$

$$Q_{P1} = 1000000$$

$$\frac{1000000}{1000000}$$

$$1.000000$$

WMI have completed

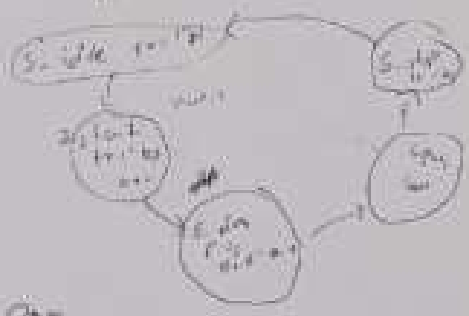
check log
data in
partially updated



not party - collecting
check - data

| | |
|-----|-----|
| 100 | 100 |
| 100 | 100 |
| 100 | 100 |
| 100 | 100 |
| 100 | 100 |

WMI - huge



Don't
Don't

seconds in

100 5/21

Rx - party - 1000 parts
Tx - party

Don't voltage

history of all voltage - 1000

↑ 3.5 prod 10 voltage

circled solution

close prod 100

Spencer

mobile data (100)

100 10
100 10

100 10



100 10

100 10 100 10

History analysis for expected data

100 10 100 10

100 10 100 10



at History analysis

Setup time

100 10 100 10

100 10 100 10

100 10

100 10 100 10



100 10 100 10

100 10 100 10

Reset / initial clk / data $\rightarrow$ history

23:06 $\rightarrow$ ISC Protocol
slave address
Timing relationship between
Read / write bit

ACK (acknowledge) bit

= 0 acknowledge (ACK)

= 1 negative acknowledge (NACK)

Data bytes $\rightarrow$ 8 bit

to allow to acknowledge

multiple data byte

stop condition

open drain

Put up resistor value
range 1k Ω - 10k Ω

Phase counter - dividing one bit into
sub periods

Phase 1: $\overline{SCL} = 0$
SCL hold

Phase 2: $\overline{SCL} = 1$
SCL = high

Phase 3: $\overline{SCL} = 1$
SCL = sample

clock constraints $\rightarrow$ timing

- make clock
- make ground clock
- set clock delay

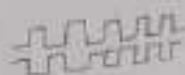
driving on both

23:06

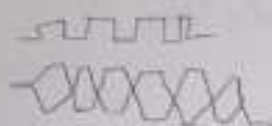
SPS Protocol

Serial 8 bit

CS, SCK, MISO, MOSI
Slave / Master (100 MHz)

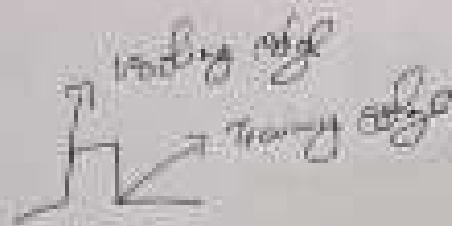


$\rightarrow$ clk polarity



$\rightarrow$ clk phase

CPOL (clock polarity)



CPOL = 0



CPOL = 1



CPHA = 0

→ falling edge

1 = Training edge

(fixed or configurable)

Combinational logic / Data chain → single or
independent slave → multiple cs

SOC → FPGA + And core

ARM → protocol → data

FPGA

ASIC protocol
training ?

Data protocol → without complexity

↳ running on basic

central → predictive / many